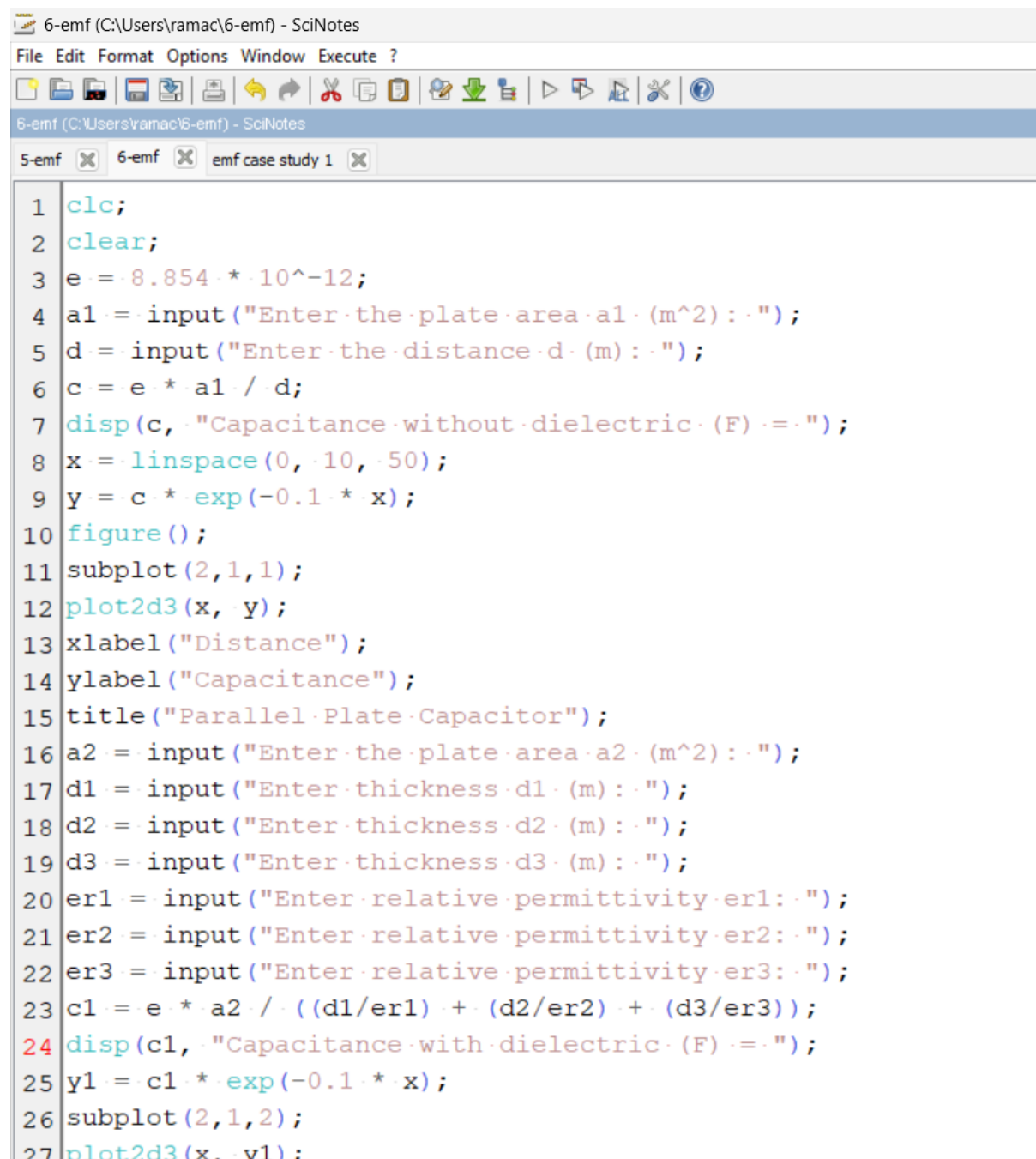


Exp no:6 Determination and plotting of Capacitance of parallel plate capacitor in three different dielectric medium with specific area, dielectric thickness and various dielectric constant.

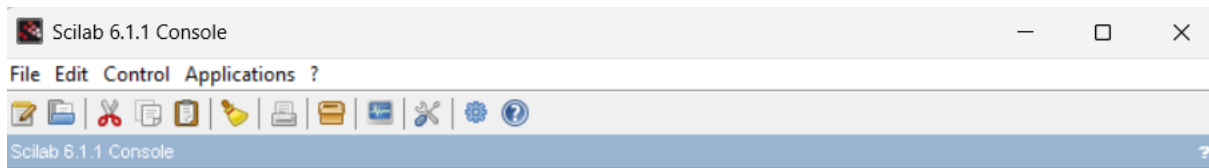
Aim: To write program for capacitance of parallel plate capacitor in three different dielectric media using SCILAB.

Software required: SCILAB version 6.0.1

Practical output:



```
6-emf (C:\Users\ramac\6-emf) - SciNotes
File Edit Format Options Window Execute ?
6-emf (C:\Users\ramac\6-emf) - SciNotes
5-emf 6-emf emf case study 1
1 clc;
2 clear;
3 e = 8.854 * 10^-12;
4 a1 = input("Enter the plate area a1 (m^2): ");
5 d = input("Enter the distance d (m): ");
6 c = e * a1 / d;
7 disp(c, "Capacitance without dielectric (F) = ");
8 x = linspace(0, 10, 50);
9 y = c * exp(-0.1 * x);
10 figure();
11 subplot(2, 1, 1);
12 plot2d3(x, y);
13 xlabel("Distance");
14 ylabel("Capacitance");
15 title("Parallel Plate Capacitor");
16 a2 = input("Enter the plate area a2 (m^2): ");
17 d1 = input("Enter thickness d1 (m): ");
18 d2 = input("Enter thickness d2 (m): ");
19 d3 = input("Enter thickness d3 (m): ");
20 er1 = input("Enter relative permittivity er1: ");
21 er2 = input("Enter relative permittivity er2: ");
22 er3 = input("Enter relative permittivity er3: ");
23 c1 = e * a2 / ((d1/er1) + (d2/er2) + (d3/er3));
24 disp(c1, "Capacitance with dielectric (F) = ");
25 y1 = c1 * exp(-0.1 * x);
26 subplot(2, 1, 2);
27 plot2d3(x, y1);
```



Enter the plate area a1 (m²): 0.0003

Enter the distance d (m): 1*10⁻³

2.656D-12

"Capacitance without dielectric (F) = "

Enter the plate area a2 (m²): 0.0005

Enter thickness d1 (m): 3*10⁻³

Enter thickness d2 (m): 4*10⁻³

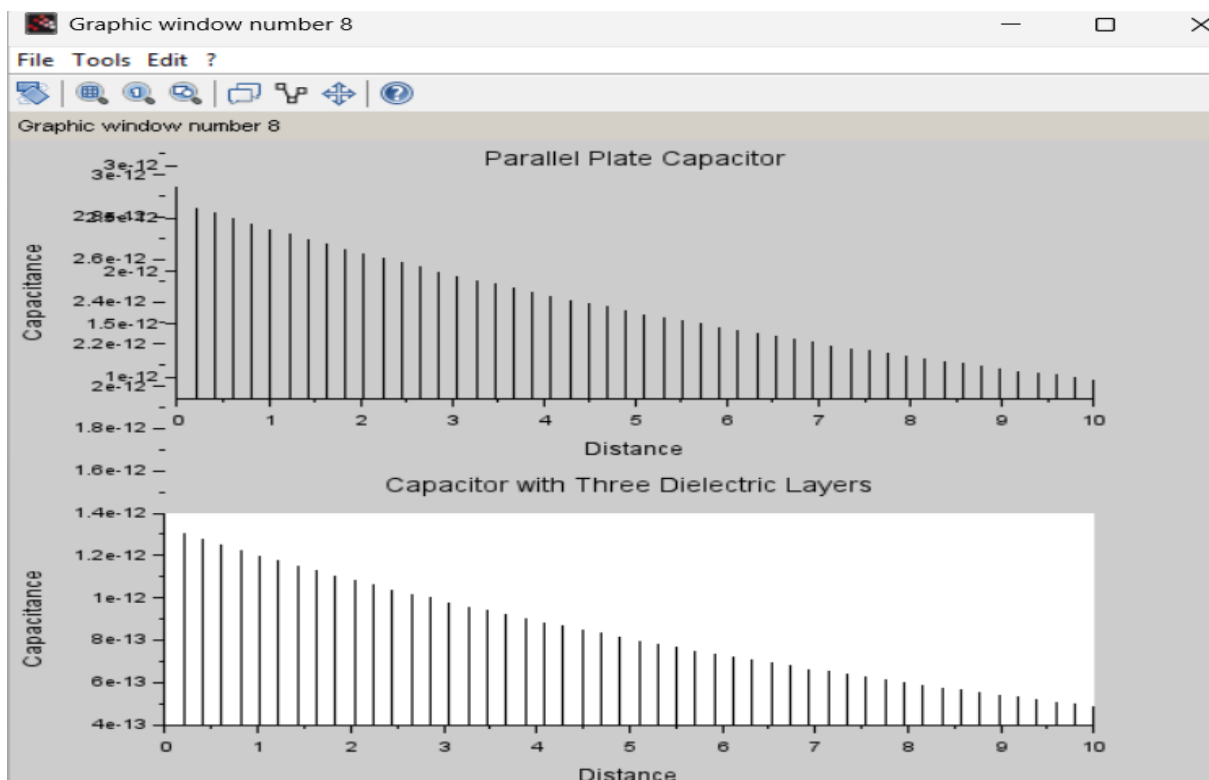
Enter thickness d3 (m): 5*10⁻³

Enter relative permittivity er1: 2

Enter relative permittivity er2: 4

Enter relative permittivity er3: 6

1.328D-12



Theoretical Output:

$\text{Area}^{(a1)} = 3 \text{ cm}^2 \Rightarrow 0.0003$
 $\text{distance } d = 1 \text{ mm}$
 $\text{Area}^{(a2)} = 5 \text{ cm}^2$
 $d_1, d_2, d_3 =$
 $\epsilon_{r1}, \epsilon_{r2}, \epsilon_{r3} =$

$\epsilon_{r1} = 2$
 $\epsilon_{r2} = 4$
 $\epsilon_{r3} = 6$

$d_1 = 3 \text{ mm} \quad 3 \times 10^{-3}$
 $d_2 = 4 \text{ mm}$
 $d_3 = 5 \text{ mm}$

① capacitance without dielectric.

$$C = \frac{\epsilon_0 A}{d}$$

$$\Rightarrow \frac{8.854 \times 10^{-12} \times 0.0003}{1 \times 10^{-3}} \Rightarrow \frac{2.656 \times 10^{-15}}{10^{-3}} \Rightarrow 2.6 \times 10^{-12}$$

② capacitance with three dielectric layers.

$$C = \frac{\epsilon_0 A}{\left(\frac{d_1}{\epsilon_{r1}} + \frac{d_2}{\epsilon_{r2}} + \frac{d_3}{\epsilon_{r3}} \right)}$$

$\frac{d_1}{\epsilon_{r1}} \Rightarrow \frac{3 \times 10^{-3}}{2} \Rightarrow 1.5 \times 10^{-3}$
 $\frac{d_2}{\epsilon_{r2}} \Rightarrow \frac{4 \times 10^{-3}}{4} \Rightarrow 1.0 \times 10^{-3}$
 $\frac{d_3}{\epsilon_{r3}} \Rightarrow \frac{5 \times 10^{-3}}{6} \Rightarrow 0.833 \times 10^{-3}$

$$\Rightarrow \frac{8.854 \times 10^{-12} \times 0.0005}{3.333 \times 10^{-3}}$$

$$\Rightarrow 1.32 \times 10^{-12} \text{ F}$$

Result : Thus the program was verified for capacitance of parallel plate capacitor in three different dielectric media using SCILAB was successfully.